Design and Development of a TCS3448-Based PAR Sensor for Agricultural Monitoring

by

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Synopsis

Precision agriculture demands real-time, spatially distributed monitoring of environmental parameters to support efficient crop management. One of the key parameters is Photosynthetically Active Radiation (PAR), which indicates the solar irradiance within the 400–700 nm waveband that drives photosynthesis. When using quantum sensors for measurement, it is usually quantified using Photosynthetic Photon Flux Density (PPFD, μmol m^{-2} s^{-1}). However, the unit cost of commercial quantum sensors capable of measuring photosynthetically active radiation with the required precision is too high, making it impractical to be widely applied on a large scale in large-scale farmlands.

This project investigates and implements a low-cost PAR sensing solution based on the TCS3448 14-channel multi-spectral sensor and the Nordic Semiconductor nRF52833 System-on-Chip. The custom firmware of TCS3448 has met the needs of register mapping, SMUX path routing, and the I²C (Inter-Integrated Circuit) bank switching mechanism.

A calibration method based on the multivariate linear regression (MLR) calibration model is applied to the eight visible-band channels (F1–F8), yielding a PAR estimate. The entire system operates within a low-power duty-cycled architecture, triggered by the nRF52833 Real-Time Clock at one-second intervals.

In addition to the electronic system, a low-cost optical diffuser was applied to provide the cosine angular response needed for reliable outdoor PAR measurement, and a housing was designed for installing the sensors and diffusion plates. Experimental angle-scan testing was employed to characterize the candidate diffuser sheet. And finally, improved MLR calibration and preliminary testing could confirm linear spectral response and acceptable PAR accuracy.

Acknowledgements

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Finally, I would like to thank the students of the Communication Engineering major. Discussing with them has always been beneficial to me.

Statement of Authorship

I, Jingjie Nan

State that this work submitted for assessment is my own and expressed in my own words. Any uses made within it of works of other authors in any form (eg. Ideas, figures, text, tables) are properly acknowledged at their point of use. A list of the references employed is included.

Date.....20/04/2026.....

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1 Introduction

1.1 Research Background and Motivation

Precision agriculture, as an advanced concept of crop management, has received widespread attention in recent years. Its core lies in precisely identifying and addressing the spatial variability within fields through modern information technologies, thereby achieving efficient resource utilization and optimized crop yields. Thanks to the rapid development of global positioning systems (GPS), wireless sensor networks, and high-performance computing capabilities, farmers can obtain real-time spatial and temporal distribution information of crops, soil, and environmental factors, and accordingly formulate differentiated management strategies. Precision agriculture is information-intensive and still has considerable room for development in the fields of sensing and mapping. It not only helps alleviate the environmental pressure, economic costs, and market competition faced by agricultural production but also significantly enhances sustainability. Stafford believes that most arable farms will fully adopt this concept to achieve precise production on a farm-wide scale.[3]

PAR determines the rate at which plants fix carbon dioxide from the atmosphere into organic matter, thereby influencing biomass accumulation, water use efficiency, and the scheduling of irrigation, fertilization, and harvesting operations. Despite its agronomic significance, real-time spatially resolved PAR monitoring remains largely absent from commercial precision agriculture deployments. Professional quantum sensors such as the LI-190R and Apogee SQ-521 offer the cosine-corrected spectral response required for accurate outdoor PPFD measurement. However, their high unit price makes it impossible to achieve a dense configuration of field instruments. A typical precision agriculture deployment may require tens to hundreds of sensing nodes distributed across a field. If commercial quantum sensors are to be used, then for every 100 m^2, one sensor node should be installed. Assuming that a PAR sensor

costs £300, the total cost for PAR monitoring of one acre (approximately 4047 m^2) of farmland would be £12141. This would be a considerable expense for agricultural workers.

Recent advances in silicon multi-spectral sensing have been producing compact and low-cost integrated circuits that provide channel-resolved photon detection across the visible spectrum. These devices offer a compelling compromise between cost, size, power consumption, and spectral fidelity [2]. The TCS3448 sensor of ams OSRAM is a 14-channel multispectral sensor that is representative of the current development level of such devices and forms the core sensing element of this project.

1.2 Project Objectives

The primary aim of this project is to design and implement a low-cost and reliable PAR sensor module for integration into multi-sensor precision agriculture nodes. This sensor module is constructed based on the Nordic Semiconductor nRF52833 system-on-chip (SoC), which features an ARM Cortex-M4F processor, a wide range of low-power peripherals, and is suitable for Bluetooth wireless integration.

The specific objectives addressed in this report are as follows. First, complete firmware was developed to interface the nRF52833 with the TCS3448 sensor over I^2C, including a full mapping of the TCS3448 register set and resolution of the significant differences between this device and its predecessor, the AS7341, from which the initial driver was adapted [1]. Second, an improved multiple linear regression calibration model was implemented in firmware to convert raw channel counts to a PAR estimate. Third, the angular cosine response of a low-cost diffuser element was characterized through simulation and experiment. Fourth, the integrated system was evaluated in indoor calibration and outdoor trial conditions with reference to another reliable PAR measurement device.

1.3 Application and Significance

The system developed in this project has direct applicability to irrigation scheduling, where PAR data can be combined with soil moisture measurements to refine evapotranspiration models. PAR time series also provide input to disease-warning systems that correlate light availability with the risk of fungal infection. At a broader scale, a network of low-cost distributed PAR sensors could provide the spatial PAR maps required to support variable-rate management of fertilizer and irrigation, reducing input waste and environmental impact. The modular design of the nRF52833 platform means that the node mentioned before is readily extensible with additional sensors, such as environment, rainfall, leaf wetness, and soil moisture modules that have been identified as complementary components within the wider project scope.

1.4 Report Structure

Chapter 2 provides the theoretical and literature background necessary to contextualize the design choices made in this project. Chapter 3 describes the overall hardware and system architecture. Chapter 4 documents the firmware

implementation in detail, including the driver porting process and calibration algorithm. Chapter 5 covers the diffuser investigation and design. Chapter 6 presents test results and performance evaluation. Chapter 7 concludes and identifies directions for future work.

2 Literature Review

2.1 Photosynthetically Active Radiation

Photosynthetically Active Radiation is defined as the component of the solar spectrum within the wavelength interval 400 - 700 nm that is absorbed by plant photosynthetic pigments, principally chlorophyll a and chlorophyll b, to drive the light-dependent reactions of photosynthesis. The internationally accepted metric for PAR in agronomic and ecological contexts is the Photosynthetic Photon Flux Density (PPFD), defined as the number of photons in the 400–700 nm band incident upon a unit surface area per unit time, expressed in micromoles of photons per square meter per second ($\mu mol \cdot m^{-2} \cdot s^{-1}$). This formulation counts photons rather than energy because of the quantum nature of the photochemical reactions, which means that a single photon drives a single primary photochemical event regardless of wavelength.

For this project, PAR was estimated using a simplified linear model in which the weighted sum of spectral channel counts from the TCS3448 sensor is mapped to PPFD through a set of regression coefficients that are derived from calibration against a reference sensor. The model takes the form:

$$PAR = b_0 + \sum_{i=1}^{n} b_i \cdot \mathrm{x_i} + \varepsilon \tag{1}$$

where b_0 is the intercept, b_i are the regression coefficients for each spectral channel, x_i are the corresponding raw channel counts, and ε is the residual. This formulation, adapted from Bäumker et al. [2], permits calibration without requiring explicit knowledge of the sensor spectral response function or the absolute irradiance of the calibration source.

2.2 Comparative Review of PAR Sensor Technology

Three principal technology paths have been pursued for PAR measurement, each offering different trade-offs between spectral accuracy, cost, and complexity.

The first candidate was professional quantum sensors, represented by the LI-190 and Apogee SQ-500 series, which employ a silicon photodiode combined with a precision optical filter that equalizes the device's spectral response across the 400 - 700 nm band to approximate the flat photon-counting response required for accurate PPFD measurement. As commercial sensors, the calibration coefficients have generally been measured at the time of manufacture. A hemispherical diffuser provides the cosine angular response needed for outdoor measurements under direct and diffuse solar radiation. These instruments achieve extremely low measurement uncertainties under calibration conditions, and are practical reference standards in agricultural research [5]. Their principal limitation for the deployment of precision agriculture is cost, which typically exceeds £300 per unit.

The second candidate was a lower-cost alternative investigated by Rajendran et al. [4], which employed a single silicon photodiode combined with a custom thin-film interference filter to approximate the PAR spectral response. This approach achieved a substantially reduced manufacturing cost, but the spectral resolution was sacrificed as the reason. The device's response is fixed by the filter design and cannot adapt to changes in the spectral composition of the incident light. Under artificial light sources, including LED grow lights common in controlled-environment agriculture, single-channel sensors of this type exhibit systematic errors that cannot be corrected without additional information.

The third candidate was the multi-channel spectral sensors, representing the most recent development in this field. Bäumker [2] demonstrated that using the eight visible band channels of an ams AS7341 spectral sensor calibrated against a professional quantum sensor with multiple linear regression could achieve a root

mean square error (RMSE) of approximately 16 $\mu mol \cdot m^{-2} \cdot s^{-1}$ across a range of natural and artificial light conditions. After investigation, this performance can be achieved with a component cost of approximately £5, which is sufficient to meet the needs of many precision agriculture applications and promote the adoption of sensor nodes.

2.3 TCS3448 Multi-Spectral Sensor

The ams-OSRAM TCS3448 is a 14-channel multi-spectral sensor implemented in a compact L3.1×W2.0×H1.0 mm OLGA package. The 14 channels comprise eight narrow-band spectral channels designated F1 through F8; three broadband channels designated FXL, FY, and FZ that allows to approximate chromaticity coordinates; a Clear channel sensitive to the full visible spectrum; a Near Infrared (NIR) channel; and a Flicker Detection (FD) channel.

The sensor incorporates a programmable analogue gain stage and six 16-bit integrating ADCs. Integration time is controlled by two registers: an 8-bit value setting the number of integration steps named ATIME, and a 16-bit value setting the duration of each step in units of 2.78 μs named ASTEP. The total integration time is given by:

$$\text{Integration time (ms)} = (\text{ATIME} + 1) \times (\text{ASTEP} + 1) \times 2.78\ \mu\text{s} \qquad (2)$$

The key architectural feature of the TCS3448 in this project is the AUTO Spectral Multiplexer (SMUX). It could route the photodiode channels' outputs onto six 16-bit ADC inputs across sequential measurement phases. Normal SMUX mapping must be configured by writing to a dedicated register bank before each measurement. To simplify firmware in this project, the auto SMUX mode 3 (enabled by the auto_SMUX bit in the CFG20 register) was implied, allowing the TCS3448 to automatically perform a three-phase acquisition with a fixed channel-to-ADC assignment, thereby reducing coding complexity, at the expense of configuration flexibility.

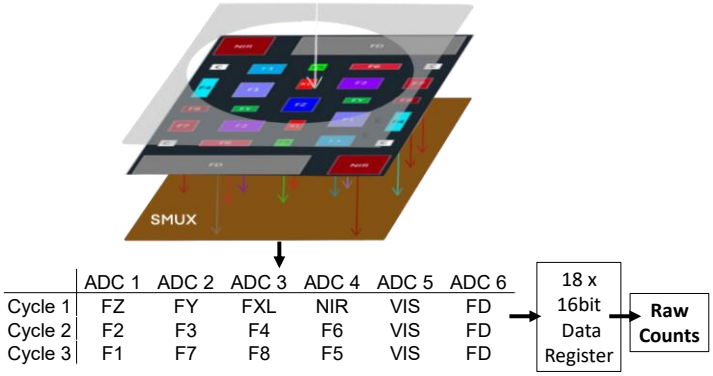


Figure 1 Schematic diagram of SMUX operation mode

2.4 Multiple Linear Regression Calibration

Multiple linear regression is a mature and effective statistical technique that can be used to construct linear prediction models for the dependent variable based on multiple independent variables. In the context of spectral PAR estimation, the independent variables are the raw channel readings from the spectral sensor and PPFD measured by the reference equipment, while the dependent variable is the predicted PAR value. The regression coefficients are determined by minimizing the average of the errors between the model-predicted values and the reference values over a dataset covering typical light conditions. This method implicitly takes into account the spectral response of the sensor and the uneven photon energy weighting across different regions in the PAR band without requiring additional effort.

The regression coefficients in this project's firmware were obtained by using the Solver function of Excel based on the raw counts under indoor lighting and outdoor natural lighting. Compared with direct regression which might calculate the negative factors, the results obtained through Solver were more in line with the physical reality.

3 System Architecture and Hardware Design

3.1 System Overview

The PAR sensing system was designed around three principal constraints: low unit cost, low power consumption, and practicality in field environments. All the architectural decisions about the hardware and firmware design were based on these constraints. A modular block decomposition was adopted to partition the system into the optical sensing subsystem, the processing subsystem, and the power subsystem.

The optical sensing subsystem captures incident radiation across the photosynthetically active spectrum (400–700 nm) and outputs raw digital counts over I²C. The processing subsystem centered on the nRF52833 acquires these counts, applies a pre-calibrated regression model, and logs the resulting PAR estimate. The power subsystem manages supply voltage regulation and enforces a low duty-cycle operating mode to extend battery life.

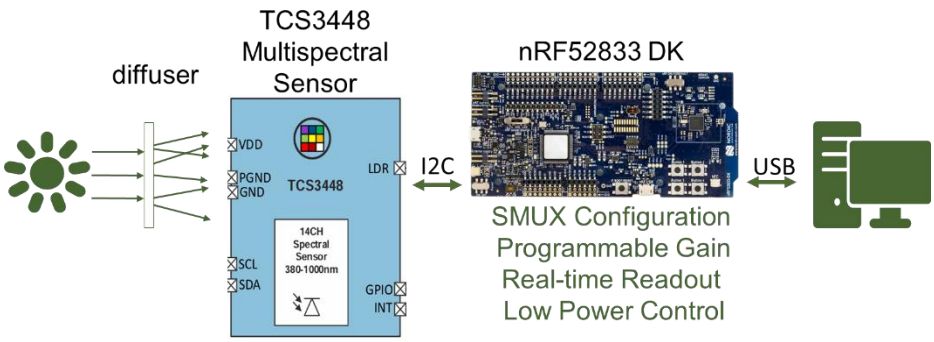


Figure 2 Simplified diagram of hardware architecture

3.2 TCS3448 and AS7341

The selection of a multi-spectral sensor is central to achieving the project's low-cost, high-accuracy objectives. During the initial evaluation phase, TCS3448 and AS7341 offered by ams-OSRAM were considered. The table below presents a quantitative comparison of key hardware parameters for both devices, compiled from their

respective official datasheets.

Table 1 Comparison of TCS3448 and AS7341's hardware

Parameter	TCS3448	AS7341	Hardware Difference
Total number of channels	14 channels (F1–F8, FXL, FY, FZ, NIR, Clear, FD)	11 channels (F1–F8, NIR, Clear, FD)	TCS3448 has 3 additional channels
Spectral range	380–1000 nm	350–1000 nm	TCS3448 has more channels concentrated in the PAR band
ADC resolution	16-bit per channel (6 independent 16-bit ADCs)	16-bit per channel (6 independent 16-bit ADCs)	Identical
VDD	1.7 V – 1.98 V	1.7 V – 2.0 V	TCS3448 has a slightly narrower range
Package	OLGA-8, 3.1 mm × 2 mm × 1 mm	OLGA-8, 3.1 mm × 2 mm × 1 mm	Identical
SMUX configuration	Supports Auto-SMUX (one-key enable via CFG20 register for automatic cycling)	Requires manual SMUX configuration	TCS3448 configuration is significantly simplified
GPIO / INT / LED driver	Fully supported	Fully supported	Identical
Typical operating current	Active: 210–280 µA Sleep: 0.7–5 µA	Active: 210–300 µA Sleep: 0.7–5 µA	TCS3448 is slightly lower
Cost	£4.33	£4.9	TCS3448 is slightly lower

Both sensors output multi-channel spectral data via an I²C interface and are suited to comparable application scenarios; however, they differ substantially in channel configuration and SMUX architecture.

All things considered, TCS3448 had a higher cost-performance ratio, could operate

for a long time with lower power consumption, and could obtain more spectral information, which proved that it's particularly suitable for agricultural sensing in this project.

3.3 nRF52833 as the Development Platform

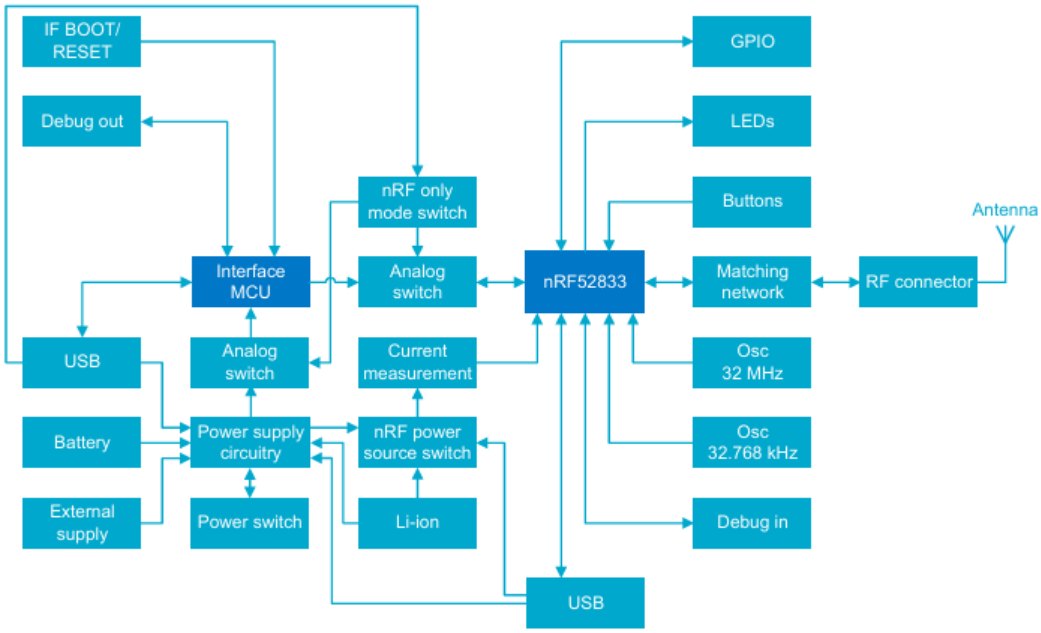


Figure 3 functional architecture of the nRF52833 DK. Source: nRF52833 DK User Guide

The nRF52833 Development Kit (DK) was selected as the host microcontroller. It integrates a 64 MHz ARM Cortex-M4 processor with a wide range of low-power peripherals. The nRF52833 includes a series of analog and digital interfaces, such as ADC, full-speed 12 Mbps USB 2.0, UART/SPI/TWI, PWM, I2S, and PDM, and has a power supply voltage range of 1.7V to 5.5V. It can be powered by rechargeable batteries or a USB for the device. It has an LDO voltage regulator and a DC-DC converter, combined with an automatic power management system, which can help achieve low power consumption even in more advanced applications. Its nRF5 SDK support provides mature, well-documented peripheral drivers, reducing the difficulty of firmware development. Furthermore, it has a low-power comparator with wake-up from System OFF mode [10], making it well-suited to duty-cycled sensing.

3.4 Power Architecture

The power subsystem, in this report especially referring to the power consumption reduction subsystem, was not confined to a specific piece of hardware but rather permeated every aspect of the entire system in practice as an important component of the project.

The nRF52833 integrates an internal DC-DC step-down converter that operates in addition to its LDO regulator. By setting the NRF_POWER->DCDCE register bit to 1 at startup, the firmware could enable the DC-DC converter for the REG1 regulator stage. This could significantly reduce power conversion losses compared to LDO-only mode when the regulator voltage drop is high. The documentation in the DevZone (the technical community of Nordic Semiconductor) reported that enabling the DC-DC converter results in as much as a 35% reduction in power usage [9], indicating this is a low-effort but high-value optimization step in the power design.

Furthermore, the system was designed to operate in a duty-cycled manner, where the nRF52833 microcontroller and the TCS3448 sensor were both optimized for low power during inactive periods. The nRF52833 microcontroller ensured that the correct voltage levels were applied for proper operation. The sensor operated at about 1.8 V, and its board-level supply was enabled by GPIO P1.08, while chip-level activity was managed through the TCS3448 ENABLE register, reducing the complexity of the firmware. If measurements are not required, the sensor will enter a low-power sleep state and will be reactivated only when needed for data collection.

The combination of these low-power configurations resulted in a power system that is well-suited for extended battery-powered deployments, reducing overall energy consumption while ensuring accurate measurements during active periods.

3.5 I²C Communication

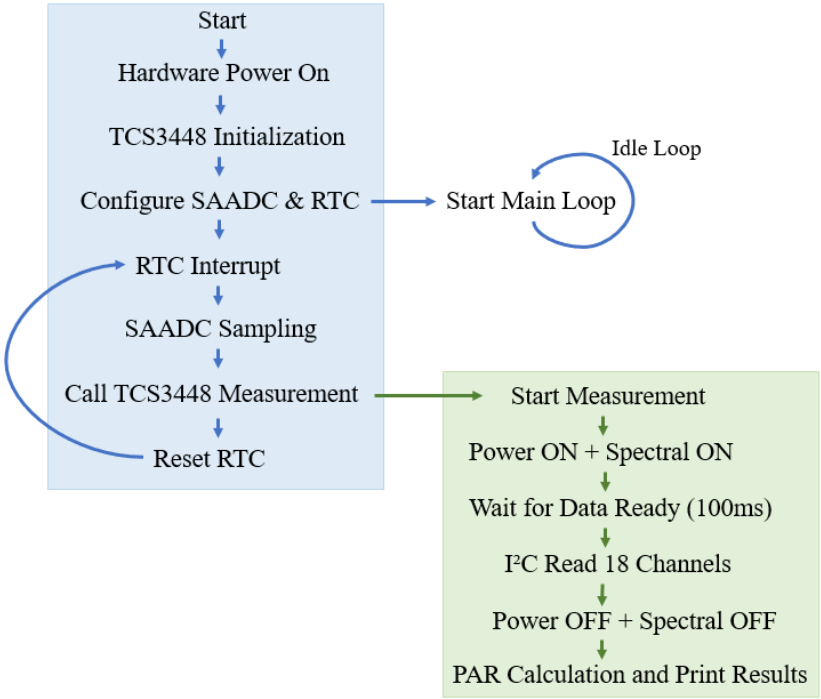
The TCS3448 sensor used the I²C protocol to communicate with the nRF52833. In

this design, the SCL (Clock) and SDA (Data) lines were connected to GPIO pins P0.17 and P0.19 on the nRF52833 respectively. These pins were used to send and receive data between the microcontroller and the sensor. This I^2C connection allowed the microcontroller to send commands to the sensor, configure settings like gain and integration time, and receive raw sensor readings from the 14 channels.

It is worth noting that in the TCS3448 data sheet, it's mentioned that the SCL and SDA interfaces require pull-up resistors. In this implementation, the internal pull-ups alone were not sufficient for reliable bus behavior, and external 4.7 kΩ pull-ups were added to improve rise time and communication stability. Due to technical issues, the influence of the other pull-up resistor value on the sensor performance has not yet been studied.

4 Firmware Design and Implementation

The design of firmware needed to satisfy the correctness of the TCS3448 register protocol, the numerical accuracy of the PAR regression model, and the low-power duty-cycled architecture suitable for field deployment, etc. This project modified the register configuration and other aspects based on the firmware of AS7341.



Block Diagram 1 Overall firmware architecture

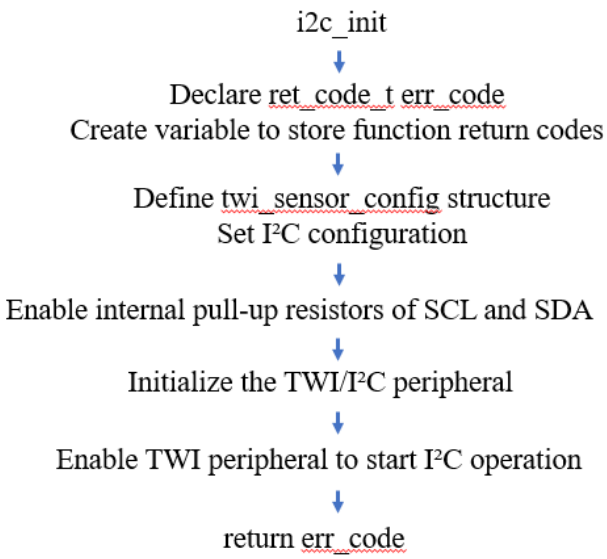
The core customizations for the TCS3448 sensor were concentrated in the TCS3448 driver files. Register configuration and device-specific definitions were primarily implemented in the tcs3448_defines.h header file. The codes in main.c and i2c_interface.c (together with i2c_interface.h) underwent only minor adaptations for system orchestration and pin mapping. In the tcs3448.c file, the majority of the changes were made to accommodate the new sensor's auto-SMUX architecture and updated channel mapping.

Functionally speaking, the i2c_interface.c file encapsulated all I²C communication functions, including read/write operations and pin configuration for the nRF52833.

The tcs3448.c file implements the complete TCS3448 sensor driver, covering device initialization, register configuration, SMUX control, power management, spectral data acquisition, and the PAR calculation algorithm that uses multiple linear regression on the eight visible channels. The main.c file manages overall system orchestration, including the initialization sequence, RTC-driven periodic sampling, sensor power enable, and low-power duty-cycled operation. The following text will provide a detailed explanation.

4.1 I²C Communication Interface (i2c_interface.c)

The i2c_interface.c file provided a clean hardware abstraction layer for TWI operations. The block diagram key initialization code is as follows (all the following diagrams have been simplified, retaining only the key parts.):



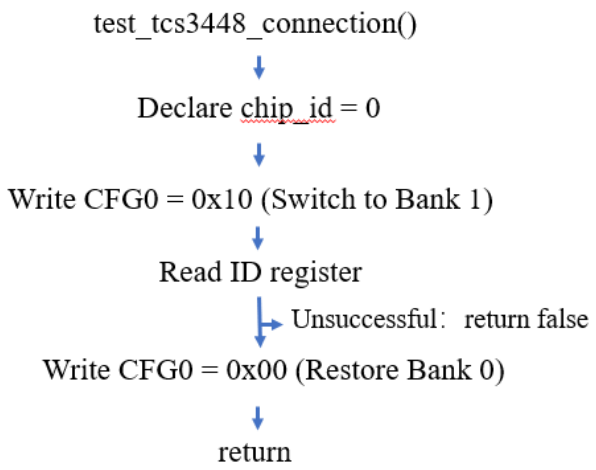
Block Diagram 2 I²C interface initialization function

In the initialization, the TWI peripheral was configured on the correct pins (P0.19 SDA, P0.17 SCL), internal pull-ups for bus robustness were enabled, and the blocking mode was used. The design kept the interface simple and synchronous as the measurement window was short, and avoided unnecessary interrupt overhead, while ensuring reliable communication with the TCS3448.

4.2 TCS3448 Sensor Driver (tcs3448.c and tcs3448_defines.h)

The majority of the project-specific modifications reside here. The registration definitions and bit masks have been updated in the tcs3448_defines.h file. All the contents were obtained from the datasheet. Due to the length of the content, no further elaboration is provided here, please refer to Appendix 3 for details.

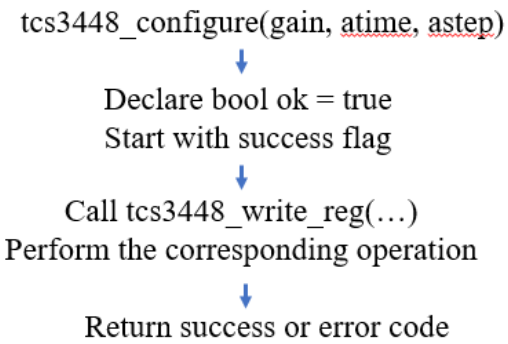
4.2.1 Device connection and bank switching:



Block Diagram 3 TCS3448 sensor connection test function

The function was switched to the auxiliary register bank, then immediately restored to bank 0 after reading the device ID (0x81). This three-step sequence is mandatory for the TCS3448 to prevent subsequent register accesses from failing.

4.2.2 Sensor configuration uses auto-SMUX mode and fixed timing:

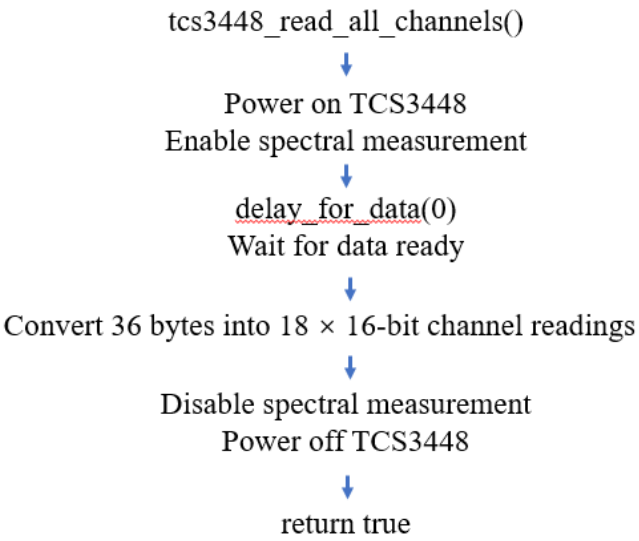


Block Diagram 4 Register configuration function

Critical registers were written sequentially to set the integration time, analogue gain, and auto-SMUX mode. The use of auto-SMUX eliminated the need to write a 20-byte routing table on every measurement, greatly simplifying the driver.

4.2.3 PAR Calculation Algorithm

First, read all 18 outputs at once:



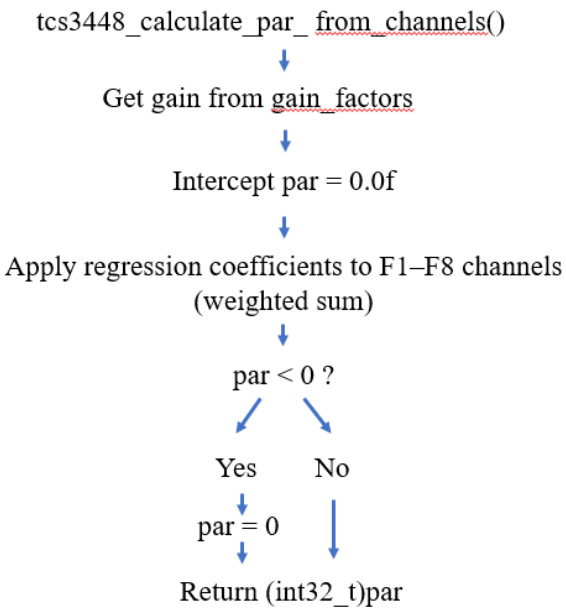
Block Diagram 5 Channel reading function

Strict power management and data readiness polling were implemented by the functions tcs3448_power_enable() and tcs3448_delay_for_data(), ensuring that the sensor operates only within the measurement window.

Then, the PAR value was computed in the same driver file.

Only the eight visible-band channels (F1–F8) were used. The calibrated regression coefficients were applied directly to raw counts. Gain and integration-time normalization were omitted in the current version because this project refers to Baumker's [2] calibration method, in which the gain and integration time have been set to fixed values.

The block diagram of the calculation part is as follows:



Block Diagram 6 PAR calculation function

4.3 System Control and Low-Power Operation (main.c)

System orchestration was handled in main.c with only minor modifications from the AS7341 baseline.

The general framework has been roughly presented at the beginning of this chapter. After enabling the DC-DC converter and sensor board power, sensor detection and initialization were then performed by the code. The RTC interrupt triggered both TCS3448 sampling and SAADC voltage monitoring. Since the TCS3448 sensor has an internal ADC, the SAADC in this firmware was not utilized for PAR measurement.

4.4 Firmware Logic Summary

The firmware followed a simple and clear execution flow as described below.

At startup, the DC-DC converter and sensor power were first enabled in main.c, then the I²C interface was initialized. It would check whether the TCS3448 sensor is connected correctly and configure it through tcs3448.c.

After initialization, the program could enter the measurement loop. Every second, the system was woken up by the Real-Time Clock (RTC). The firmware then turned on

the sensor, started reading all 14 spectral channels, calculated the PAR value using the regression algorithm inside tcs3448.c, and turned the sensor off again.

This repeating cycle was initialized once, then measured once per second with the sensor powered only when needed, keeping the overall power consumption low while ensuring reliable PAR data collection. The three main files could work together smoothly as the timing and power were controlled by main.c, communication is handled by i2c_interface.c, and sensor and PAR calculation were managed by tcs3448.c.

Ideally, according to the data sheet, the power consumption would be as follows:

Table 2 Power consumption estimation

Component	**State**	**Current (approx.)**	**Duty Fraction**	**Average Current**
nRF52833 CPU	Active (64 MHz, DCDC)	2.6 µA [11]	1000/1000	2.6µA
TCS3448	Active	210~280 µA [7]	100/1000	21~28 µA
TCS3448	Sleep	0.7~5 µA [7]	900/1000	0.63~3.5 µA
Total (estimated)	/	/	/	24.23~34.1 µA

This estimation was idealized. The actual power consumption would vary depending on the task load, sampling frequency of the sensors, power mode adopted, etc. Actually, the system was duty-cycled at the application level, but the true power consumption performance generally does not reach such a low level.

5 Diffuser and Calibration

5.1 Role and Requirements of the Diffuser

Accurate outdoor measurement of PAR requires the sensor to exhibit a cosine angular response. This means the output must be directly proportional to the cosine of the angle of incidence θ between the incoming ray and the surface normal:

$$R_\theta = R_0 cos\theta \qquad (3)$$

where R_0 is the response at normal incidence (θ=0), R_θ is the response when the incidence angle is θ. Without cosine correction, direct-beam sunlight at large zenith angles and diffuse skylight produce substantial measurement errors. Professional quantum sensors such as the LI-190R achieve this response through a precision hemispherical diffuser combined with an optical filter.

In this project, a simple flat diffuser plate needed to be placed above the TCS3448 sensor to approximate the required cosine response. The diffuser must maintain high transmittance in the 400–700 nm PAR band and provide sufficient scattering while keeping component cost below £1.

By conducting a survey of the common diffuser sheets available in the market, it is found that various products, such as IBROLIGHT's BIM-6320-01 Cosine Corrector and LBTEK's 200 nm - 1100 nm Cosine Corrector, choose to use 1mm PTFE as the diffuser sheet. Furthermore, PTFE has excellent transmittance and near-ideal scattering properties, and its long-term outdoor durability makes it highly suitable for this project.

5.2 Angle-Scan Testing

The performance target adopted was the minimum scatter recommendation curve published by ams-OSRAM in the TCS3448 datasheet.

To quantify the cosine response, a simple angle scanning testing device was

constructed, consisting of a sensor at a fixed position and a light source that moves around the sensor as the center. Under controlled indoor conditions, measurements were conducted from -90° to +90° with increments of 10°.

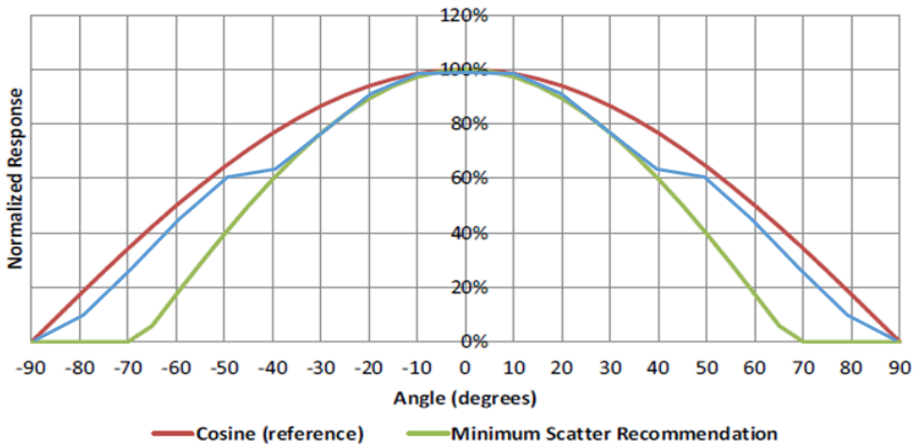


Figure 4 The measured response of the 1 mm PTFE diffuser (blue curve)

Table 3 The measured response of the 1 mm PTFE diffuser (Un-calibrated)

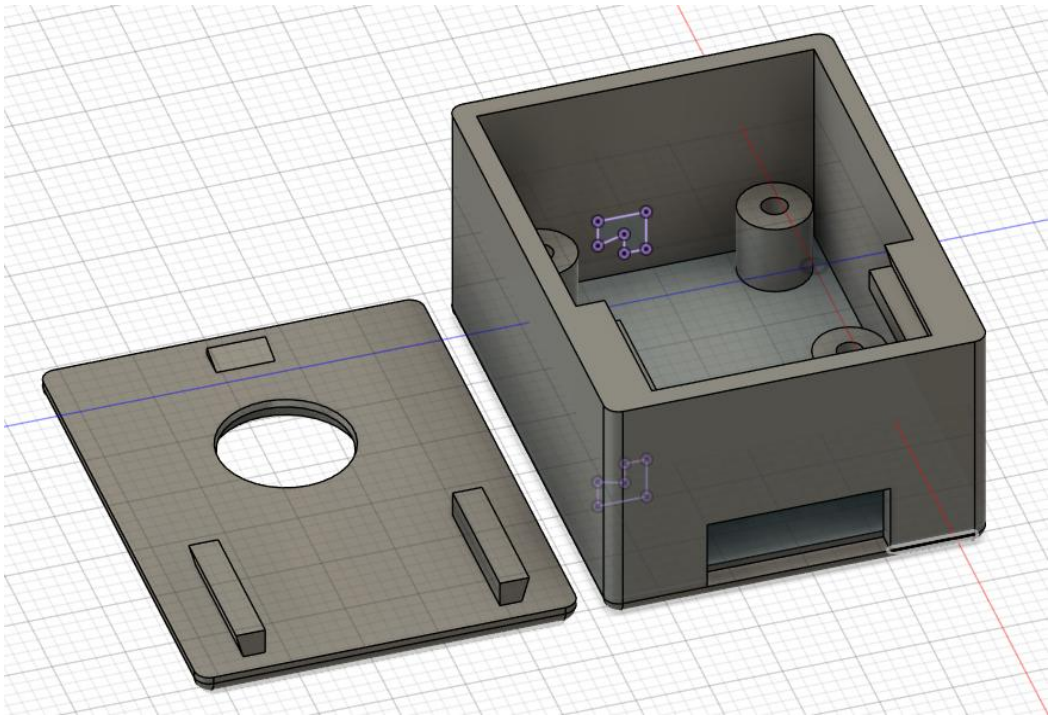
degree	0	10	20	30	40	50	60	70	80	90
PAR	178	177	164	139	115	110	82	50	20	4
Percentage	100%	99%	92%	78%	65%	62%	46%	28%	11%	2%
Error (%)	0	0.97	1.95	9.83	15.7	3.86	7.87	17.9	35.3	/

Figure 4 shows the measured response of the 1 mm PTFE diffuser (blue curve) overlaid on the ideal cosine response (red) and the minimum scatter recommendation (green). The PTFE diffuser closely followed the cosine curve up to ±70° with a maximum deviation of less than 10 %. In fact, it was observed that the curve is not very smooth. This might be due to the insufficient accuracy of the manual operation of the measuring equipment.

5.3 Housing Design

A custom housing was designed to securely hold the TCS3448 sensor board and the PTFE diffuser plate, providing them with the necessary protection. It is 3D-printed in PLA using the parameters shown in the figure below to balance strength and weight, as well as to reduce the print time.

The diffuser was placed on a 1mm-deep socket on the lid. The lid is securely connected to the box using mortise and tenon joints, ensuring that the lid would not accidentally fall off under normal conditions. There was a hole on the bottom side of the box, which allowed the DuPont wires connecting the sensor and the development board to pass through. Overall dimensions were 34 × 26× 18 mm, resulting in a compact, low-profile component suitable for canopy or ground-level deployment.



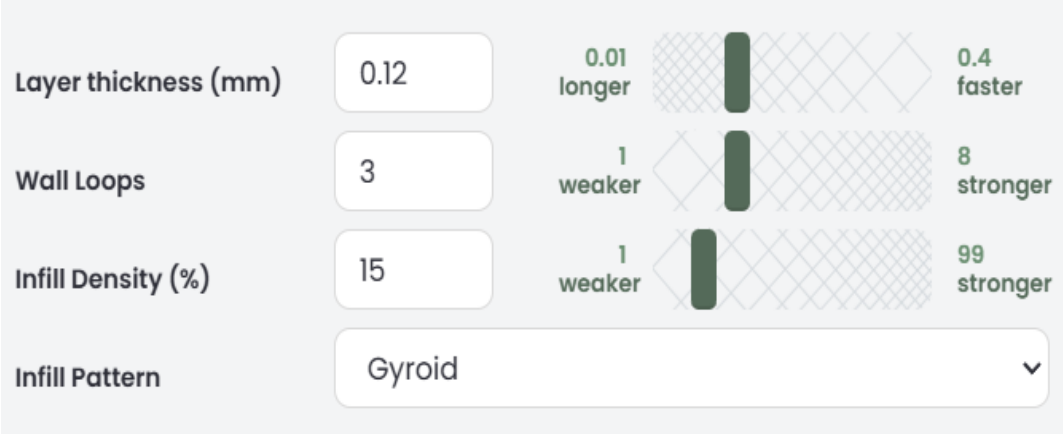


Figure 5 Modeling and 3D Printing

5.4 Sensor Calibration

5.4.1 Non-negative optimization of Regression Calibration

This project has made further improvements based on the method of obtaining parameters through the direct application of the regression operation in MLR.

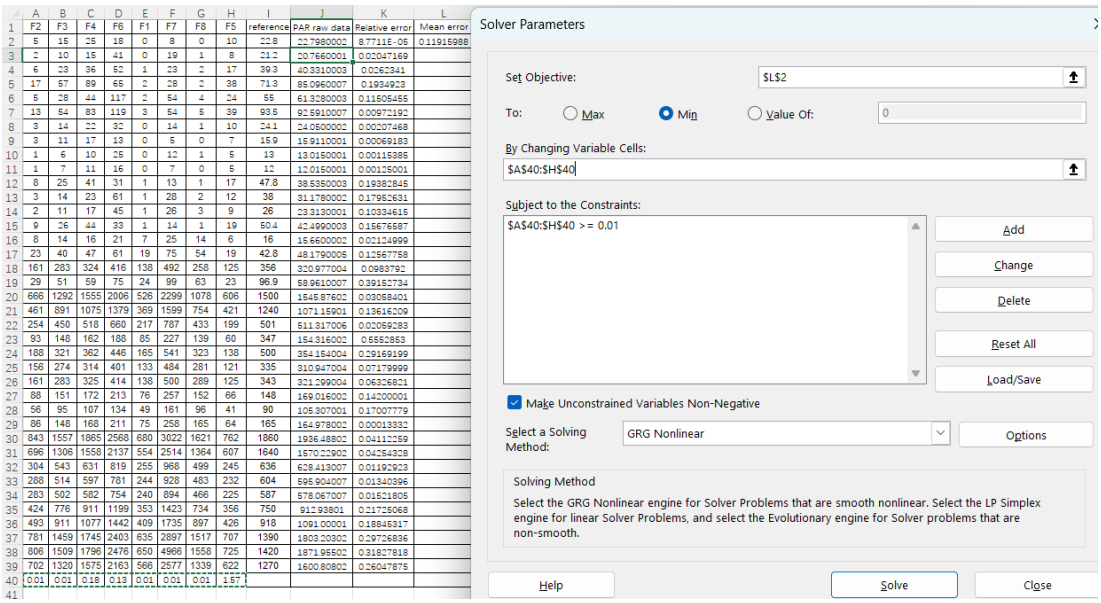


Figure 6 Excel Solver Parameter Configuration

In Excel, all relevant channel readings for the eight visible bands (F1–F8) were first entered. The regression coefficients for these channels were initially set to the minimum allowable value of 0.01. Using these coefficients, the initial predicted PAR value was calculated for each calibration sample. The predicted PAR was then compared with the corresponding reference PPFD value from the quantum sensor, and the relative error for each sample was computed using the formula:

$$Relative\ error = \frac{TCS3448\ PAR\ raw\ data - reference\ PAR}{reference\ PAR} \tag{4}$$

In Excel Solver, the parameters were configured so that the eight channel coefficients became the variable cells. The objective was set to minimize the mean relative error, subject to the constraint that each coefficient must be greater than or equal to 0.01. After solving, the channel coefficients were obtained.

Instead of using the regression algorithm of Excel directly, the non-negative optimization of the relative error constraint to obtain physically meaningful positive coefficients.

5.4.2 Specific Method

Referring to the MLR formula in Chapter 2:$PAR = b_0 + \sum_{i=1}^{n} b_i \cdot \mathrm{x_i} + \varepsilon$.

In this project, x_i represented the raw counts of the output from channels F1 to F8. During calibration, b_i was initially set to 1 for all values, so as to obtain the unprocessed data. The residuals ε were used to calculate the mean relative errors for regression coefficients.

Calibration data were collected under two representative illumination conditions, indoor artificial lighting and outdoor natural sunlight. A total of 38 measurements was recorded simultaneously with the TCS3448 sensor and a reference detector.

The final coefficients implemented in firmware are as follows:

Table 4 Final regression coefficients

F2	F3	F4	F6	F1	F7	F8	F5
0.01	0.01	0.18373	0.12553	0.01	0.01	0.01	1.56495

These values successfully produced a linear response across the full PAR range of interest and were preliminarily validated in both indoor and preliminary outdoor trials.

The diffuser investigation and housing design described above, together with the MLR calibration procedure, completed the hardware front-end of the low-cost PAR sensor. The resulting system has met the cosine-response requirement for outdoor use while maintaining component cost well below that of commercial quantum sensors.

6 Result analysis

This chapter presents the experimental results obtained from the TCS3448-based PAR sensing system, including indoor spectral response and outdoor trial measurements.

6.1 Spectral Channel Response

Following successful firmware deployment, the TCS3448 was operated under LED lighting to check the results. As can be seen from the screenshot, the distance between the test light source and the sensor gradually increased from close to far, and the measurement results also decreased linearly with the distance.

```
Debug Terminal

<info> app: Integration time = 100.08 ms
<info> app: TCS3448 Par: 804
<info> app: Sample: 118 (103 mV)

<info> app: Integration time = 100.08 ms
<info> app: TCS3448 Par: 636
<info> app: Sample: 118 (103 mV)

<info> app: Integration time = 100.08 ms
<info> app: TCS3448 Par: 372
<info> app: Sample: 120 (105 mV)

<info> app: Integration time = 100.08 ms
<info> app: TCS3448 Par: 190
<info> app: Sample: 118 (103 mV)
```

Figure 7 Screenshot of PAR measurement results

During the testing process, it can be observed that the original output values of the F1 and F8 channels under LED were extremely small. However, under natural light, the situation where individual channels have extremely small output values does not usually occur (refer to Appendix 2). This situation was in line with the laws of reality. The F1 spectrum is close to ultraviolet light, the F8 spectrum is close to infrared light, while the LED spectrum is concentrated in the range of red to blue light, so it is difficult to be detected and output by the sensor. The output values of the outdoor sunlight path were relatively balanced because sunlight encompasses all measurable spectral ranges. This has further confirmed that the measurement results of the sensor

were in line with the actual situation, thus making its spectral channel response reliable.

6.2 MLR Calibration Performance

Bäumker used the direct regression method when calibrating the AS7341 sensor. Due to the limited number and type of samples collected for this project, the accuracy of the calibrated sensor was not perfect. To achieve better accuracy, some minor upgrades have been made to the regression method.

In this program, the regression coefficients derived using the non-negative optimization of relative error of Excel Solver were embedded in firmware and applied to the raw counts in real time. The resulting PAR estimates were compared against the reference device measurements over the combined indoor and outdoor dataset. The scatter plot of TCS3448's estimated PAR versus reference PPFD values is presented in Figure 8. The relationship is near-linear across the full measurement range with no systematic curvature or step-change visible.

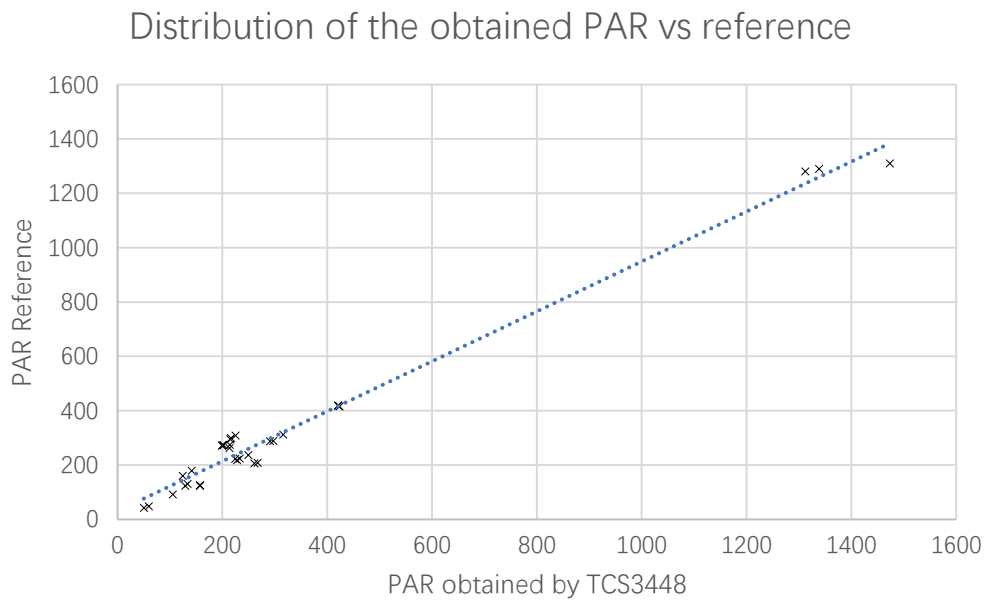


Figure 8 Scatter plot of TCS3448 PAR versus reference PPFD (μmol m^{-2} s^{-1}). The dashed line represents the trendline, which is close to 1:1.

6.3 Result statistics

The experimental results were recorded, and the Root Mean Square Error (RMSE) was calculated separately for the measurement results only under neutral light, only under indoor light, and under all light conditions.

Table 5 Result statistics and residual comparison

		TCS3448	Reference	Residual	RMSE
Indoor	Neutral light	421	418	3	
		420	419	1	
		291	287	4	
		224	222	2	
		316	312	4	
		50	42.3	7.7	
		129	123	6	
		59	48.1	10.9	
		133	131	2	
		105	91.5	13.5	
		233	224	9	
		249	237	12	
		297	288	9	
		228	218	10	
		423	416	7	7.751774
	Cold light	200	273	-73	
		141	179	-38	
		199	271	-72	
		124	160	-36	
		202	273	-71	
	Warm light	157	124	33	
		267	208	59	
		157	126	31	
		261	206	55	34.0045
Outdoor	Shadow	215	294	-79	
		216	299	-83	
		225	309	-84	
		213	262	-49	
		214	274	-60	
	Sunlight	1312	1280	32	
		1338	1290	48	42.98025

Overall, the TCS3448 sensor performed well under neutral light conditions with minimal errors, indicating its reliability in everyday environments. The outdoor measurement under strong light results also showed relatively good performance.

However, in cold light, warm light, and under outdoor shade, the sensor's performance declined with larger errors. Bäumker's report also addressed the same issue of decreased accuracy after changing the type of light source. This may reflect that the current dataset sample is still insufficient. To obtain more accurate results in different environments, the spectral response of the sensor needs to be optimized, especially for responses to different light sources.

6.4 Appearance of the finished product

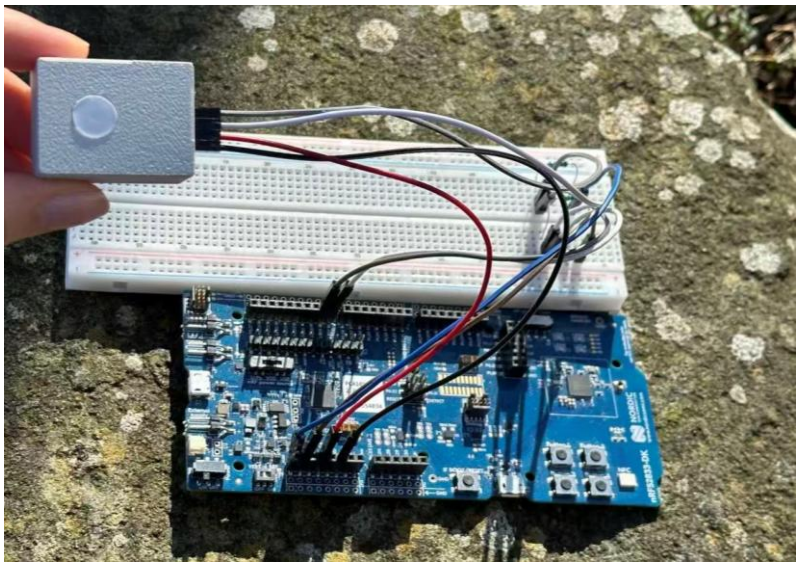


Figure 9 Hardware real photo

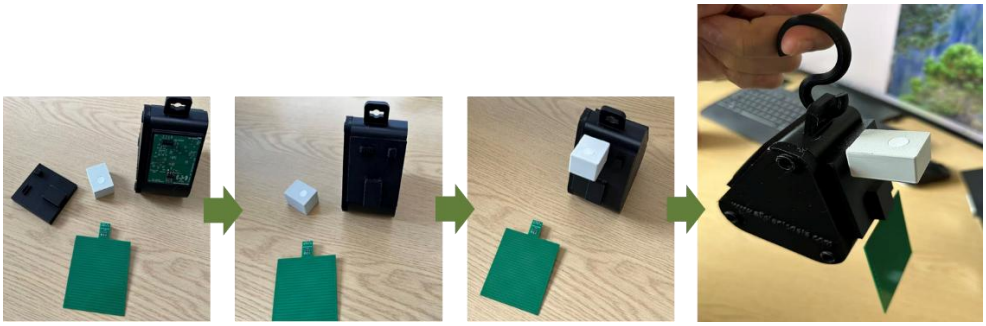


Figure 10 Finished product image of multi-sensor node

7 Risk Assessment and Mitigation

Table 6 Risk Matrix

Likelihood \ Impact	Negligible	Minor	Moderate	Major	Catastrophic
Rare	Low	Low	Low	Medium	High
Unlikely	Low	Low	Medium	Medium	High
Possible	Low	Medium	Medium	High	High
Likely	Medium	Medium	High	High	High
Almost Certain	Medium	High	High	High	High

The following is the estimation and evaluation of the main risks.

Table 7 Risk Register

Risk Description	Likelihood	Impact	Mitigation Measures	Residual Risk Level
Incorrect register mapping or SMUX configuration (TCS3448 vs AS7341 differences) leading to invalid channel data or failed auto-SMUX.	Possible	Major	Full datasheet comparison; ID_REG test in test_tcs3448_connection(); extensive logging; unit tests on all 14 channels.	Low
I^2C communication instability (pull-up resistors, voltage mismatch at 1.8 V).	Possible	Moderate	4.7 kΩ external pull-ups implemented; internal pull-ups enabled in code; error handling and retry logic in i2c_interface.c.	Low
MLR calibration coefficients fail to generalize to outdoor spectral conditions (negative coefficients or high RMSE).	Possible	Major	Indoor & outdoor calibration dataset; validation against reference PAR meter; non-negative regression algorithm; par_regression_coeffs array validated.	Medium

Low-cost PTFE diffuser fails to provide acceptable cosine angular response for outdoor PAR measurement.	Possible	Major	Angle-scan laboratory testing vs reference cosine curve; iterative material/thickness selection (1 mm PTFE selected); 3D housing design with diffuser slot.	Low
Actual power consumption exceeds low-power target (duty-cycled RTC at 1 s intervals), shortening battery life.	Unlikely	Moderate	DC-DC converter enabled (NRF_POWER->DCDCEN = 1); sleep modes; power gating of TCS3448 via tcs3448_power_enable().	Low
Moisture, dust, or UV damage to electronics or diffuser during outdoor field trials.	Likely	Major	Sealed 3D-printed housing (Gyroid infill, 0.12 mm layer); conformal coating option; weather monitoring during trials.	Medium
Safety (3D Printing / Materials)	Unlikely	Minor	Use of ventilated printer area; PPE (gloves, mask); material safety data sheets followed.	Low
Incorrect PAR values leading to misleading precision agriculture recommendations.	Possible	Major	Cross-validation with commercial sensor; raw channel logging; error bounds in firmware; clear documentation of calibration conditions.	Medium
Data Management and Cyber Security	Unlikely	Minor	Sensor output logs stored locally; back up periodically to prevent data loss.	Low

The majority of risks are technical in nature and have already been substantially mitigated through careful firmware design, hardware choices and experimental preliminary validation. With the controls outlined above, the overall project risk profile is Low to Medium.

8 Conclusions and Discussion

This chapter evaluates the outcomes of the project against the specific objectives established in Chapter 1, reflects on the principal technical challenges encountered and their resolutions, characterizes the limitations of the delivered system, and identifies concrete directions for future development.

8.1 Achievement of Project Objectives

1. TCS3448 Firmware and Register Mapping:

A complete I2C driver was implemented for the TCS3448, including the dual-bank register switching, auto-SMUX configuration via CFG20, and a 36-byte burst read of all 18 channel outputs. The three critical differences from the predecessor AS7341, including bank switching protocol, SMUX architecture, and channel count, were systematically resolved. The firmware correctly recovered all 14 spectral channels on every measurement cycle with no data corruption observed in testing.

2. Multiple Linear Regression PAR Calibration:

A non-negative optimization regression model was calibrated using 38 measurements across indoor LED and outdoor natural light conditions.The constraint to non-negative coefficients is essential in this part. Unconstrained ordinary method produced negative regression coefficients that generated unphysical negative PAR estimates under low-light conditions, a problem that the Solver-constrained approach entirely eliminated.

3. Diffuser Cosine Response Characterization:

Achieved within the stated performance target. The 1 mm PTFE diffuser was evaluated over the 90° angular range. From 0° to 70°, the measured response deviated from the ideal cosine by a maximum of 17.9 %. Beyond 70°, the response fell more steeply, reaching only 11 % of the normal-incidence value at 80° compared

to the ideal cosine prediction of 17 % but remaining within the minimum scatter recommendation published in the TCS3448 datasheet.

4. Indoor and Outdoor Evaluation:

Indoor and outdoor tests verify a linear spectral response across the full sensor dynamic range. However, the evaluation dataset is limited in scope as it does not include measurements under artificial grow-light spectra, which are increasingly common in controlled-environment agriculture, nor does it include extended multi-day logging to assess long-term stability. The final eight-channel model yielded an RMSE of approximately 40 μmol m^{-2} s^{-1} against the reference sensor. As the sample size is small, the RMSE was larger than the 16 μmol m^{-2} s^{-1} reported by Bäumker et al., but its performance was still much better than that obtained by directly using regression. It is worth noting that if measurements are only conducted under indoor LED neutral light, the RMSE would be as small as only 7.75μmol m^{-2} s^{-1}, which is an impressive result. From this, it can be speculated that the new calibration method has potential, but more samples need to be collected and further verification is still required. These gaps represent the primary unresolved element of the fourth objective.

8.2 Technical Problems and Discussions

1. I2C Bus Instability

Early development was impeded by the I2C non-acknowledgement from the TCS3448. The data indicated that the built-in pull-up resistors in SCL and SDA have excessively high resistance values, which may cause the pull-up time to be too long and thereby affect the measurement. Adding 4.7 kΩ external pull-up resistors to both SDA and SCL reduced the rise time and eliminated all communication failures.

2. SMUX Configuration Complexity

The AS7341 driver used as the starting point required explicit 20-byte SMUX routing table writes before each measurement. Porting this approach to the TCS3448 produced inconsistent channel assignments because the routing table structure differs between devices. Enabling the TCS3448's auto-SMUX mode (CFG20 = 0x60) resolved this completely. The device could automatically cycle through its three fixed measurement phases and deliver all 14 channels without firmware-side routing management. This architectural difference between the two sensors was not apparent from a cursory reading of the TCS3448 datasheet; it required careful comparison of the CFG20 register descriptions. Therefore, the lesson is that driver porting between sensor generations always requires a full register-level audit, not a superficial pin and address update.

3. Regression Coefficients Problems

The initial calibration attempt used the standard Excel Regression function, which produced negative coefficients for some channels, and the measured values had a large error compared to the reference PAR meter. Physically, a negative coefficient implies that higher photon counts in a visible-band channel reduce the estimated PPFD, which is impossible under the assumed linear model. Replacing Regression with a Solver-based non-negative formulation resolved the issue while improving the physical interpretability of the model.

8.3 Limitations

1. Calibration Dataset Scope

The 38-measurement calibration dataset, although sufficient to produce a workable regression model, is still insufficiently diverse to guarantee generalization across all practical deployment scenarios. The dataset does not include measurements under monochromatic LED grow-light arrays, which exhibit narrow-band spectral power distributions that fall outside the training distribution. Applying the current

regression coefficients to such light sources is likely to introduce systematic errors that cannot be detected without reference measurements. More measurements are still required for a robust model.

2. Sensor Saturation at High Irradiance

At full summer midday irradiance (PPFD > 1800 μmol m^{-2} s^{-1}), channels may saturate at the 16-bit ADC ceiling of 65535 counts with the current gain setting of 4×. Saturation in any input channel invalidates the regression model, because the mapping from counts to PPFD is no longer linear. The fixed gain and integration time significantly reduce the flexibility of the system. If it is to be truly deployed, an adaptive algorithm needs to be used.

3. Manual Angle-Scan Testing Accuracy

Angle-Scan Testing is performed by manually rotating the light source relative to the sensor in 10° increments. Figure 4 reveals a sudden change in the cosine response at 40-50°. A motorized rotation stage with an angular resolution of 1° or better would substantially improve characterization accuracy.

8.4 Directions for Future Work

1. Adaptive Gain Control

Implementing an automatic gain step-down algorithm triggered by channel saturation detection is the highest-priority firmware improvement. This change requires no hardware modification and would extend the reliable measurement range to full sunlight without user intervention.

2. Bluetooth Low Energy Data Transmission

The nRF52833 SoC integrates a Bluetooth function that has not been activated in this project. Enabling BLE advertisement of PAR data would allow the node to participate in a multi-node wireless sensor network without additional hardware.

3. Integration with the Multi-Sensor Node

The PAR module is designed as one component of a broader multi-sensor node that also includes soil moisture, environmental, and leaf wetness sensors. Integrating these sensors onto a shared I2C bus with the TCS3448, verifying electrical compatibility, and preliminarily validating the combined power budget against the duty-cycle design established in Chapter 4 represent the next system-level development milestones.

4. Long-Term Outdoor Stability

A continuous outdoor logging trial with concurrent measurements from the reference sensor at one-hour intervals would quantify sensor drift, the durability of the PTFE diffuser under UV exposure, and the weather-tightness of the 3D-printed housing. Data from this trial would also provide the richer dataset required for the improved calibration described above.

8.4 Self-Reflection and Review

This project provided direct experience with the full hardware-to-firmware development cycle on a real embedded platform. At the outset, the assumption was that adapting an existing AS7341 driver to the TCS3448 would be a straightforward register substitution. The experience proved otherwise: the register bank switching mechanism, auto-SMUX architecture, and changed channel-to-output mapping each required independent investigation and debugging. This taught me that documentation must be read in its entirety before writing any code, and that assumption-based porting is a significant source of subtle bugs in embedded development.

Working through the calibration problem has deepened my understanding of applied statistics in engineering. The Solver-based constrained optimization approach

required independent reading and is now a practical tool I would reach for in future data-driven projects.

The diffuser characterization work required skills at the boundary of optics, mechanical design, and experimental methodology that were new to me. During the design process of the 3D-printed model of the sensor housing, a lot of practical engineering knowledge was also acquired, which will be beneficial in the future.

8.5 Conclusion

This project has delivered a functional low-cost PAR sensing module based on the TCS3448 14-channel multi-spectral sensor and the nRF52833 SoC. The firmware correctly acquired all 14 spectral channels via the auto-SMUX mechanism, applied a non-negative regression model to the eight visible-band channels, and logged the resulting PAR estimate at one-second intervals within a duty-cycled low-power architecture. The 1 mm PTFE diffuser provided an acceptable cosine angular response to 70° at a component cost well below £1. The total estimated system power consumption of 24–34 μA was suitable for multi-month battery operation.

Every problem had a defined technical pathway to resolution. Together, these findings confirmed that the low-cost multi-spectral approach to PAR sensing is technically viable for precision agriculture deployment.

Reference

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Appendix 1: Meeting Records

Meeting 1: 26/09/2025

Agenda: Professor gave an overview of the entire project.

Action Item: Students are required to independently search for the relevant information of the sensors they are responsible for and submit it.

Meeting 2: 04/10/2025

Agenda: Students received the development board and other hardware required for the project.

Action Item: Student needed to complete environment configuration and test development board, initially getting familiar with the use of sensors.

Meeting 3: 17/10/2025

Agenda: Progress report presentation.

Action Item: Student began to write the TCS3448 code and managed to successfully run.

Meeting 4: 07/11/2025

Agenda: Professor provided the TCS3448 sensor

Action Item: Students need to test and correct the code.

Meeting 5: 14/11/2025

Agenda: Discussion and Resolution of Code Issues

Action Item: Student attempt to solve the output problem after getting a new sensor.

Meeting 6: 12/01/2026

Agenda: Progress report and code optimization

Action Item: After confirming the accuracy of the PAR value, student proceed to the next stage and investigate the relevant knowledge of the diffuser.

Meeting 7: 05/02/2026

Agenda: Progress report and 3D printed shell guidance

Action Item: Student design the shell model on software Fusion 360 and then use 3D printer to print it.

Meeting 8: 26/02/2026

Agenda: Code organization and discussion on running results

Action Item: Student further adjust the design of the shell model and observe the influence of the diffuser sheet on the measurement results.

Meeting 9: 10/03/2026

Agenda: Calibration method investigation and report

Action Item: Student further improve the code and prepare to conduct calibration. At the same time, continue to prepare the content for the poster display.

Meeting 10: 07/04/2026

Agenda: Hardware return and report guidance

Action Item: Student collect all the data and continue writing the final report.

Appendix 2: Calibration form

F2	F3	F4	F6	F1	F7	F8	F5	reference	PAR raw data	Relative error
5	15	25	18	0	8	0	10	22.8	22.79800019	8.77112E-05
2	10	15	41	0	19	1	8	21.2	20.76600012	0.020471692
6	23	36	52	1	23	2	17	39.3	40.33100028	0.026234104
17	57	89	65	2	28	2	38	71.3	85.0960007	0.193492296
5	28	44	117	2	54	4	24	55	61.32800033	0.115054551
13	54	83	119	3	54	5	39	93.5	92.59100065	0.009721918
3	14	22	32	0	14	1	10	24.1	24.05000017	0.002074682
3	11	17	13	0	5	0	7	15.9	15.91100013	0.000691832
1	6	10	25	0	12	1	5	13	13.01500007	0.001153852
1	7	11	16	0	7	0	5	12	12.01500008	0.001250007
8	25	41	31	1	13	1	17	47.8	38.53500031	0.193828445
3	14	23	61	1	28	2	12	38	31.17800017	0.179526311
2	11	17	45	1	26	3	9	26	23.31300013	0.103346149
9	26	44	33	1	14	1	19	50.4	42.49900032	0.156765867
8	14	16	21	7	25	14	6	16	15.66000018	0.021249989
23	40	47	61	19	75	54	19	42.8	48.17900052	0.125677582
161	283	324	416	138	492	258	125	356	320.9770037	0.098379203
29	51	59	75	24	99	63	23	96.9	58.96100067	0.391527341
666	1292	1555	2006	526	2299	1078	606	1500	1545.876017	0.030584011
461	891	1075	1379	369	1599	754	421	1240	1071.159012	0.136162087
254	450	518	660	217	787	433	199	501	511.3170059	0.020592826
93	148	162	188	85	227	139	60	347	154.316002	0.555285297
188	321	362	446	165	541	323	138	500	354.1540042	0.291691992
156	274	314	401	133	484	281	121	335	310.9470036	0.071799989
161	283	325	414	138	500	289	125	343	321.2990037	0.063268211
88	151	172	213	76	257	152	66	148	169.016002	0.142000013
56	95	107	134	49	161	96	41	90	105.3070012	0.170077792
86	148	168	211	75	258	165	64	165	164.9780019	0.000133322
843	1557	1865	2568	680	3022	1621	762	1860	1936.48802	0.041122592
696	1306	1558	2137	554	2514	1364	607	1640	1570.229017	0.042543282
304	543	631	819	255	968	499	245	636	628.4130071	0.011929234
288	514	597	781	244	928	483	232	604	595.9040067	0.013403962
283	502	582	754	240	894	466	225	587	578.0670066	0.015218047
424	776	911	1199	353	1423	734	356	750	912.9380101	0.21725068
493	911	1077	1442	409	1735	897	426	918	1091.000012	0.188453172
781	1459	1745	2403	635	2897	1517	707	1390	1803.203019	0.297268359
806	1509	1796	2476	650	4966	1558	725	1420	1871.95502	0.318278183
702	1320	1575	2163	566	2577	1339	622	1270	1600.808017	0.260478754
0.01	0.01	0.18	0.13	0.01	0.01	0.01	1.565			

Appendix 3 tcs3448_defines.h

```
// I2C Address / Identification
#define TCS3448_I2CADDR_DEFAULT 0x59 ///< TCS3448 default 7-bit I2C addr
#define TCS3448_CHIP_ID         0x81 ///< Expected value read from ID reg (0x5A)
#define TCS3448_AUXID    0x58 ///< AUXID register (aux identification)
#define TCS3448_REVID    0x59 ///< REVID register (revision ID)
#define TCS3448_ID_REG   0x5A ///< ID register (returns 0x81)

// Core configuration / timing / threshold
#define TCS3448_ENABLE      0x80 ///< Main enable: PON, ALS_EN, FDEN, WEN, etc.
#define TCS3448_ATIME       0x81 ///< ALS integration time cycles (ATIME)
#define TCS3448_WTIME       0x83 ///< Wait time between measurement cycles

#define TCS3448_ALS_TH_L_L  0x84 ///< ALS low-threshold LSB for interrupt
#define TCS3448_ALS_TH_L_H  0x85 ///< ALS low-threshold MSB
#define TCS3448_ALS_TH_H_L  0x86 ///< ALS high-threshold LSB
#define TCS3448_ALS_TH_H_H  0x87 ///< ALS high-threshold MSB

#define TCS3448_CFG10       0x65 ///< Flicker detect persistence FD_PERS[2:0]
#define TCS3448_CFG12       0x66 ///< ALS_TH_CH[2:0] => which channel drives thresholds/persistence

//Integration step size ASTEP works together with ATIME.
//tint(ms) = (ATIME+1) * (ASTEP+1) * 2.78µs (same style as AS7341).

#define TCS3448_ASTEP_L     0xD4 ///< Integration step size LSB (ASTEP[7:0])
#define TCS3448_ASTEP_H     0xD5 ///< Integration step size MSB (ASTEP[15:8])

// Auto-zero / gain ceiling / auto-SMUX / flicker timing
#define TCS3448_AZ_CONFIG   0xDE ///< Auto-zero frequency for ALS engines
#define TCS3448_AGC_GAIN_MAX 0xD7 ///< Max AGC gain for flicker detect, etc.
#define TCS3448_CFG8        0xC9 ///< FIFO_TH, etc. (FIFO threshold config)
#define TCS3448_CFG20       0xD6 ///< auto_smux mode / FD_FIFO_8b
#define TCS3448_FD_TIME1    0xE0 ///< Flicker detection integration timing LSB
#define TCS3448_FD_TIME2    0xE2 ///< Flicker detection timing MSB + gain
#define TCS3448_FD_CFG0     0xDF ///< FIFO write control for flicker data

// Status / interrupt / ID
#define TCS3448_STATUS2     0x90 ///< AVALID, ASAT_DIGITAL/ANALOG, FDSAT_*
#define TCS3448_STATUS3     0x91 ///< INT_ALS_H / INT_ALS_L
#define TCS3448_STATUS      0x93 ///< Primary status: ASAT, AINT, FINT, SINT
#define TCS3448_ASTATUS     0x94 ///< "ALS status" snapshot; also can be FIFO'ed
                                 ///< (ASTATUS is written to FIFO if enabled)
#define TCS3448_STATUS5     0xBB ///< SINT_FD / SINT_SMUX (system int source)
#define TCS3448_STATUS4     0xBC ///< FIFO_OV, OVTEMP, FD_TRIG, ALS_TRIG,
                                 ///< SAI ACTIVE, INT BUSY
#define TCS3448_ALS_INT_HIGH_MSK 0b00100000 ///< INT_ALS_H (high threshold crossed)
#define TCS3448_ALS_INT_LOW_MSK  0b00010000 ///< INT_ALS_L (low threshold crossed)

// ADC result registers
#define TCS3448_CH0_DATA_L  0x95 ///< ADC CH0 data low byte
#define TCS3448_CH0_DATA_H  0x96 ///< ADC CH0 data high byte
#define TCS3448_CH1_DATA_L  0x97 ///< ADC CH1 data low byte
#define TCS3448_CH1_DATA_H  0x98 ///< ADC CH1 data high byte
#define TCS3448_CH2_DATA_L  0x99 ///< ADC CH2 data low byte
#define TCS3448_CH2_DATA_H  0x9A ///< ADC CH2 data high byte
#define TCS3448_CH3_DATA_L  0x9B ///< ADC CH3 data low byte
#define TCS3448_CH3_DATA_H  0x9C ///< ADC CH3 data high byte
#define TCS3448_CH4_DATA_L  0x9D ///< ADC CH4 data low byte
#define TCS3448_CH4_DATA_H  0x9E ///< ADC CH4 data high byte
#define TCS3448_CH5_DATA_L  0x9F ///< ADC CH5 data low byte
#define TCS3448_CH5_DATA_H  0xA0 ///< ADC CH5 data high byte
#define TCS3448_CH6_DATA_L  0xA1 ///< ADC CH6 data low byte
#define TCS3448_CH6_DATA_H  0xA2 ///< ADC CH6 data high byte
#define TCS3448_CH7_DATA_L  0xA3 ///< ADC CH7 data low byte
#define TCS3448_CH7_DATA_H  0xA4 ///< ADC CH7 data high byte

// Misc config blocks (banked / misc control)
#define TCS3448_CFG0        0xBF ///< LOW_POWER, REG_BANK, WLONG, etc.
#define TCS3448_CFG1        0xC6 ///< AGAIN[4:0] (ALS gain control 0.5x..2048x)
#define TCS3448_CFG3        0xC7 ///< SAI (sleep-after-interrupt)
#define TCS3448_CFG6        0xF5 ///< SMUX_CMD[4:3] (SMUX command control)
#define TCS3448_CFG8        0xC9 ///< FIFO_TH etc.
#define TCS3448_CFG9        0xCA ///< SIEN_FD / SIEN_SMUX system int enables
#define TCS3448_CFG12       0x66 ///< ALS_TH_CH channel select
#define TCS3448_CFG20       0xD6 ///< auto_smux mode select / FD_FIFO_8b

#define TCS3448_PERS        0xCF ///< APERS[3:0]: ALS interrupt persistence
#define TCS3448_GPIO        0x6B ///< GPIO polarity/dir/in/out
#define TCS3448_LED         0xCD ///< LED_ACT / LED_DRIVE[6:0] current limit
#define TCS3448_FD_STATUS   0xE3 ///< Flicker Detect status bits (FD_VALID...)
#define TCS3448_INTENAB     0xF9 ///< Interrupt enable: ASIEN / ALS_IEN / FIEN / SIEN
#define TCS3448_CONTROL     0xFA ///< SW_RESET, ALS_MAN_AZ, FIFO_CLR, CLEAR_SAI_ACT

#define TCS3448_FIFO_MAP    0xFC ///< Which CHx / ASTATUS / FD data go into FIFO
#define TCS3448_FIFO_LVL    0xFD ///< FIFO fill level (# of 2-byte entries)
#define TCS3448_FDATA_L     0xFE ///< FIFO data LSB (auto-increment readout)
#define TCS3448_FDATA_H     0xFF ///< FIFO data MSB
```